GOVERNANCE · RISK · TECHNOLOGY ADVISORY

Côte d'Ivoire Mining Sector: Governance Gaps, Risk & AI Monitoring Roadmap

A standalone advisory note for the Ministry of Mines, DGMG and SODEMI. Identifies regulatory gaps, benchmarks infrastructure against Ghana, Mali, Mauritania and Liberia, and recommends two best-fit technology options for each priority area — with phased implementation, indicative ROI and a 2026–2030 roadmap.

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1. Executive Summary

This advisory note supplements the *Côte d'Ivoire Mining & Minerals Sector Note* (October 2026) by addressing four questions raised by the Ministry of Mines: (i) what regulatory gaps remain in the current framework, (ii) how the country's infrastructure compares with competitor jurisdictions in West Africa, (iii) what ESG and community risks require active management, and (iv) which advanced software, monitoring and artificial-intelligence systems the Government can deploy to strengthen regulation, capture revenue and improve processes. The note is structured as a standalone advisory that can be read on its own or alongside the prior sector note for fuller context.

Three findings dominate. First, the regulatory framework — anchored in the 2014 Mining Code — has at least seven material gaps: the dis-application of fiscal-stability guarantees, the absence of closure bonding methodology, the thin formalisation regime for artisanal mining, the convention-level (rather than codified) treatment of local content, the voluntary status of tailings management standards, the partial transposition of the UEMOA Community Code, and the candidate-status (rather than compliant-status) EITI position with no beneficial ownership disclosure. Each of these is addressable through a combination of regulatory reform and technology deployment.

Second, the infrastructure deficit — particularly rail and dedicated mineral-port capacity — is the binding constraint on bulk-commodity growth (manganese, nickel, iron ore) and a structural disadvantage against Ghana, Mauritania and Liberia for those commodities. Côte d'Ivoire remains competitive for gold and high-value cargoes, but the diversification strategy will not materialise without infrastructure investment that no single project sponsor is willing to underwrite alone.

Third, two best-fit technology options exist for each priority area. For regulatory compliance: a digital closure-bond calculator with bank guarantees and a UBO register integrated with EITI reporting (with optional graph-AI for ownership-network analysis). For infrastructure monitoring: IoT weigh-in-motion sensors at mine gates and satellite-based stockpile and shipment monitoring. For ASM formalisation: satellite + AI illegal-mining detection, and a mobile-based ASM registration and gold traceability platform tied to the SIMEP refinery. For ESG: IoT geotechnical sensors on tailings dams, and a multi-sensor water-quality network for the Bia/Tanoé cross-border system with Ghana. Process-improvement technologies — RPA, predictive analytics, digital twin, generative AI and an open data portal — apply across all use cases.

The recommended roadmap is phased over 2026–2030. Phase 1 (2026) builds foundations: the National Mining Data Platform, an upgraded e-cadastre portal, an ASM satellite-detection pilot in two regions, and the design of a UBO register. Phase 2 (2027–28) scales to full national coverage of ASM detection, brings the UBO register live, deploys tailings IoT sensors on priority dams, and integrates SIMEP gold traceability. Phase 3 (2029–30) layers advanced analytics — AI fiscal anomaly detection, generative AI for policy briefings, a digital twin for mine-site and infrastructure corridor planning, and an open data portal for transparency. Indicative capex across the full stack is in the USD 10–25 million range with payback periods of one to three years for the highest-impact components.

2. Regulatory Gaps

Côte d'Ivoire's 2014 Mining Code (Law No. 2014-138 of 24 March 2014) and its implementing decree (2014-97) provide the foundation of the regulatory framework, supplemented by the UEMOA Community Mining Code (Regulation 02/2023/CM/UEMOA), the Environment Code (1996), the Water Code (1998), the Rural Land Code (1998), the Forestry Code (2014), the 2018 Investment Code and the Kimberley Process certificate regime. While the framework has been broadly stable since 2014, the 2025 finance law demonstrated that stability guarantees are partial, and the ongoing code-revision process (announced in July 2024) is the opportunity to close seven specific gaps that materially affect investor predictability, fiscal capture and ESG performance.

2.1 Fiscal Stability Guarantee — Partial Dis-application

Article 164 of the 2014 Code provides a fiscal stability guarantee that historically protected operators from adverse changes to the royalty and tax regime for the life of the mining convention. The 2025 finance law raised the gold ad valorem tax by two percentage points and explicitly dis-applied Article 164 for those revisions. Perseus Mining reported paying approximately USD 20 million in additional royalties, including amounts for earlier periods. The gap is therefore definitional: a stability guarantee that can be overridden by ordinary finance law is not a stability guarantee in the sense investors understand. The risk implication is that operators will price a higher discount rate into Ivorian projects relative to jurisdictions with constitutional-level stability protection, and that future finance laws can layer further increases without convention renegotiation.

2.2 Closure Bonding Methodology — Absent

The 2014 Code requires mine-closure planning and environmental rehabilitation but does not specify a bonding methodology, calculation basis or independent surety mechanism. In practice, closure bonding is negotiated at the convention level, leading to inconsistency across operators and exposing the State to contingent liability if an operator becomes insolvent or abandons a site. Comparable jurisdictions (Ghana, South Africa) require risk-based closure bonds calculated by independent third parties and held in escrow or as bank guarantees. The risk implication is that the State may inherit unfunded closure liabilities — particularly relevant for the first-wave mines approaching end-of-life in the 2030s.

2.3 ASM Formalisation Framework — Thin

The national artisanal mining rationalisation programme has established 100 local technical committees between 2014 and 2024, but there is no national licensing register for artisanal operators, no production declaration regime, and no chain-of-custody for ASM gold between pit and the proposed SIMEP refinery. The Kimberley Process framework covers diamonds but not gold. The result is that significant gold production — estimates vary widely but a 5–15 tonne-per-year informal flow is plausible — sits outside the formal fiscal and statistical system. The risk implication is dual: revenue leakage to the State and ESG externalities (mercury use, deforestation, river pollution) that cannot be attributed or remediated.

2.4 Local Content — Convention-Level Only

A GIZ-funded 2022 assessment found that the national employment target had been exceeded, with 94% of mining-sector employees Ivorian by 2019, but gaps remain in supply-chain and subcontracting localisation. Local content requirements are currently applied at the convention level rather than codified in the Mining Code itself, leading to inconsistency across operators and weak enforceability. The revised code is expected to formalise local content rules — but the gap persists until enactment. The risk implication is that supply-chain value capture remains below potential, and that operators face unequal obligations depending on the negotiating strength of their conventions.

2.5 Tailings Management and Critical Minerals — Voluntary

The Global Industry Standard on Tailings Management (GISTM) is not codified into Ivorian law; adoption is voluntary and varies by operator. The 2014 Code does not include a specific regime for critical minerals (lithium, cobalt, rare earths, coltan), which are currently governed by the general mining-title regime despite their distinct environmental and supply-chain characteristics. The risk implication is that as critical-minerals exploration accelerates (permits granted from 2024 onward), the State lacks the technical regime to manage their specific risks, and that a tailings-dam failure would expose the voluntary-standards gap in a high-visibility way.

2.6 UEMOA Community Code — Partial Transposition

The UEMOA Community Mining Code (Regulation 02/2023/CM/UEMOA) provides a regional layer on transparency, local content and environmental standards. Côte d'Ivoire has partially transposed the regulation but has not aligned all national provisions, particularly on contract disclosure and beneficial-ownership transparency. The risk implication is that the State is exposed to regional peer review and potential UEMOA enforcement, and that investors face inconsistency between national and regional requirements.

2.7 EITI and Beneficial Ownership — Candidate Status

Côte d'Ivoire is an EITI candidate country but has not yet achieved compliant status. Beneficial ownership disclosure is not mandated, leaving the State without visibility into the ultimate owners of mining permits — a gap that has facilitated corruption and illicit financial flows in comparable jurisdictions. The 2023 EITI Standard requires beneficial ownership disclosure by 2024 for compliant countries. The risk implication is reputational (EITI suspension or de-listing), fiscal (inability to verify royalty payments against actual production), and governance (potential for politically exposed persons to hold permits undisclosed).

Table 2.1 — Summary of Regulatory Gaps and Risk Implications

#	Gap	Current State	Risk Implication	Priority
2.1	Fiscal stability guarantee	Art. 164 dis-applied by 2025 finance law	Higher discount rates; future fiscal layering	High
2.2	Closure bonding methodology	Negotiated at convention level; no codified basis	Unfunded closure liability for State	High
2.3	ASM formalisation framework	100 local committees; no licensing register	5–15 t/yr informal gold flow; ESG externalities	Critical
2.4	Local content codification	Convention-level only; not in Mining Code	Supply-chain value leakage; inconsistent obligations	Medium
2.5	Tailings & critical minerals	GISTM voluntary; no critical-minerals regime	Tailings failure exposure; unmanaged critical risks	High
2.6	UEMOA Community Code transposition	Partial transposition; contract disclosure weak	Regional peer review; investor inconsistency	Medium
2.7	EITI & beneficial ownership	Candidate status; no UBO disclosure	Reputational, fiscal and governance risks	High

Table 2.1: Seven regulatory gaps identified, with risk implication and priority rating.

3. Infrastructure Gap Analysis vs Competitor Jurisdictions

Infrastructure is the most under-discussed constraint on Côte d'Ivoire's mining diversification ambitions. Gold mines operate within the existing logistics envelope because gold doré is a high-value, low-volume cargo. But the country's manganese, nickel and iron-ore growth strategy depends on power, rail and port capacity that does not yet exist at the required scale, and the gap between policy aspiration and infrastructure reality is the binding constraint on bulk-commodity economics over the next decade. This chapter benchmarks Côte d'Ivoire against Ghana, Mali, Mauritania, Liberia and Burkina Faso on five infrastructure dimensions and identifies where the gaps are binding.

3.1 Power Supply

Côte d'Ivoire has one of West Africa's more developed power sectors, with installed capacity above 2,200 MW and a generation mix dominated by gas-fired thermal (CIPREL, Azito, CIRÉ) and hydroelectricity (Kossou, Taabo, Buyo). The country is a net electricity exporter to Ghana, Burkina Faso, Mali and Guinea. Power tariffs for industrial users are broadly competitive within West Africa, although reliability remains a constraint in the more remote northern regions where several new gold projects (Koné, Séguéla) are located. For manganese and nickel, power costs matter more: both are bulk-processed (manganese ore is washed and screened; nickel laterite can be processed via high-pressure acid leach or smelting), and the energy intensity of those processes means power tariffs directly determine project economics. Ghana offers comparable industrial tariffs and slightly better reliability; Mali and Burkina Faso are weaker on both dimensions. The implication is that Côte d'Ivoire is competitive on power for gold and high-value cargoes, but a step-change in available power at competitive tariffs is required to make local processing of manganese and nickel economically feasible.

3.2 Rail — The Critical Missing Infrastructure

Rail is the single most important infrastructure gap. The existing Abidjan-Ouagadougou railway (Sitarail) runs through the centre of the country toward Burkina Faso but does not extend to the western iron-ore belt or the northern manganese zone. Multiple feasibility studies have proposed heavy-haul extensions to the western iron-ore deposits — Mount Klahoyo and Segaye — but no construction has started, and the capital cost (estimated in excess of USD 2 billion for a dedicated heavy-haul line) has deterred financiers. The comparison with Mauritania and Liberia is instructive: Mauritania's SNIM railway has moved iron ore from Zouerate to Nouadhibou since 1963 and underpins one of Africa's largest iron-ore export operations; Liberia's Yekepa-Buchanan railway (originally Lamco, now ArcelorMittal) is the analogous infrastructure that enables Nimba iron-ore exports. Without a comparable rail link, Côte d'Ivoire's iron-ore resource (estimated at 4,000 Mt) is effectively stranded, and the manganese sector's growth ceiling is set by trucking economics rather than resource availability.

3.3 Ports — Abidjan and San Pedro

Côte d'Ivoire has two deepwater ports: Abidjan (the principal commercial port, expanded in 2022 with the second container terminal Vridi 2 to roughly 2.5 million TEU capacity) and San Pedro (the second port serving the western and south-western regions, expanded in 2021 with doubled capacity). Mining exports — gold doré, manganese ore and nickel — move through Abidjan, and the port's capacity is sufficient for current volumes but will require further expansion if iron-ore exports ever materialise. San Pedro is the logical export route for any future western iron-ore operation, provided a rail link is built. The comparison with Ghana (Tema and Takoradi, with dedicated mineral export facilities) and Senegal (Dakar, which serves Mali's gold exports via rail) shows that Côte d'Ivoire's port capacity is adequate for gold and manganese at current volumes, but a dedicated bulk-handling facility at San Pedro is a prerequisite for iron-ore development.

3.4 Roads — Adequate but Constrained for Bulk Commodities

The road network connecting mining regions to Abidjan has been progressively upgraded, but sections remain bottlenecks, particularly during the rainy season. The Abidjan-Bouaké-Korhogo corridor serves the northern gold and manganese mines; the Abidjan-Man corridor serves the western gold belt around Ity. Road haulage costs are competitive within the region but constitute a significant share of operating costs for bulk commodities — for manganese, road transport to port can account for 30–40% of total delivered cost, which is the structural reason why manganese economics are sensitive to ore grade. Côte d'Ivoire's paved-road density is comparable to Ghana and superior to Burkina Faso and Mali, but the absence of rail means that roads carry load that would be more economically moved by rail in any bulk-commodity scenario.

3.5 Water and Tailings Infrastructure

Water availability is a localised rather than national constraint. The central gold belt around Yaouré and Bonikro draws from the Bandama river system, and the northern mines rely on local water management infrastructure that has been sufficient for current operations. Tailings management has become more visible following the global tailings standards revision (GISTM) and EITI-aligned disclosure requirements. Most operating mines have conformant tailings storage facilities, but closure bonding (see §2.2) and the absence of a tailings-specific monitoring regime are gaps that technology can address (Chapter 9).

Table 3.1 — Comparative Infrastructure Position vs Competitor Jurisdictions

Dimension	Côte d'Ivoire	Ghana	Mali	Mauritania	Liberia	Burkina Faso
Installed power (MW, 2024)	~2,200	~5,200	~900	~880	~225	~600
Industrial tariff (USD/kWh, approx.)	0.10–0.13	0.11–0.14	0.13–0.16	0.12–0.15	0.20+	0.15–0.18
Heavy-haul rail to mines	None (Sitarail passenger/freight only)	Limited (Takoradi–Awaso bauxite)	None	SNIM Zouerate–Nouadhibou	Yekepa–Buchanan (Arcelor Mittal)	None
Deepwater ports	2 (Abidjan, San Pedro)	2 (Tema, Takoradi)	None (landlocked)	1 (Nouadhibou)	1 (Buchanan, plus Monrovia)	None (landlocked)
Paved-road density (km/100 km^2)	~7.5	~7.0	~3.5	~1.0	~3.0	~4.5
Mineral export handling	Adequate (gold/manganese); none (iron ore)	Established (gold, bauxite, manganese)	Road to Dakar/Abidjan	Iron-ore dedicated	Iron-ore dedicated	Road to Abidjan/Tema
Comparative advantage	Strong for gold; weak for bulk	Strong overall	Weak (landlocked, rail absent)	Strong for iron ore	Strong for iron ore	Weak (landlocked)

Table 3.1: Comparative infrastructure benchmark across six West and West-Central African mining jurisdictions. Sources: World Bank, IFC, national grid operators, company filings.

4. Risk, ESG & Community

This chapter consolidates the top ten sector risks, examines the ESG dimensions of the Ivorian mining sector, and identifies the community and land-tenure issues that require active management. The risk matrix below maps likelihood against severity for the ten risks identified across the prior sector note and the gaps in Chapter 2. The ESG analysis then covers the environmental, social and governance dimensions in turn, with cross-border externalities discussed at the end.

4.1 Risk Matrix — Top 10 Sector Risks

The matrix below scores the ten most material risks on a 1–5 likelihood scale (1 = rare, 5 = almost certain) and a 1–5 severity scale (1 = minor, 5 = severe). The scoring is illustrative — based on the analysis in this advisory note — and should be treated as a starting point for a formal Ministry risk register maintained on a quarterly basis. The risks cluster into three categories: fiscal-regulatory (R1, R4), operational (R3, R6, R7, R8, R10) and social-political (R2, R5, R9).

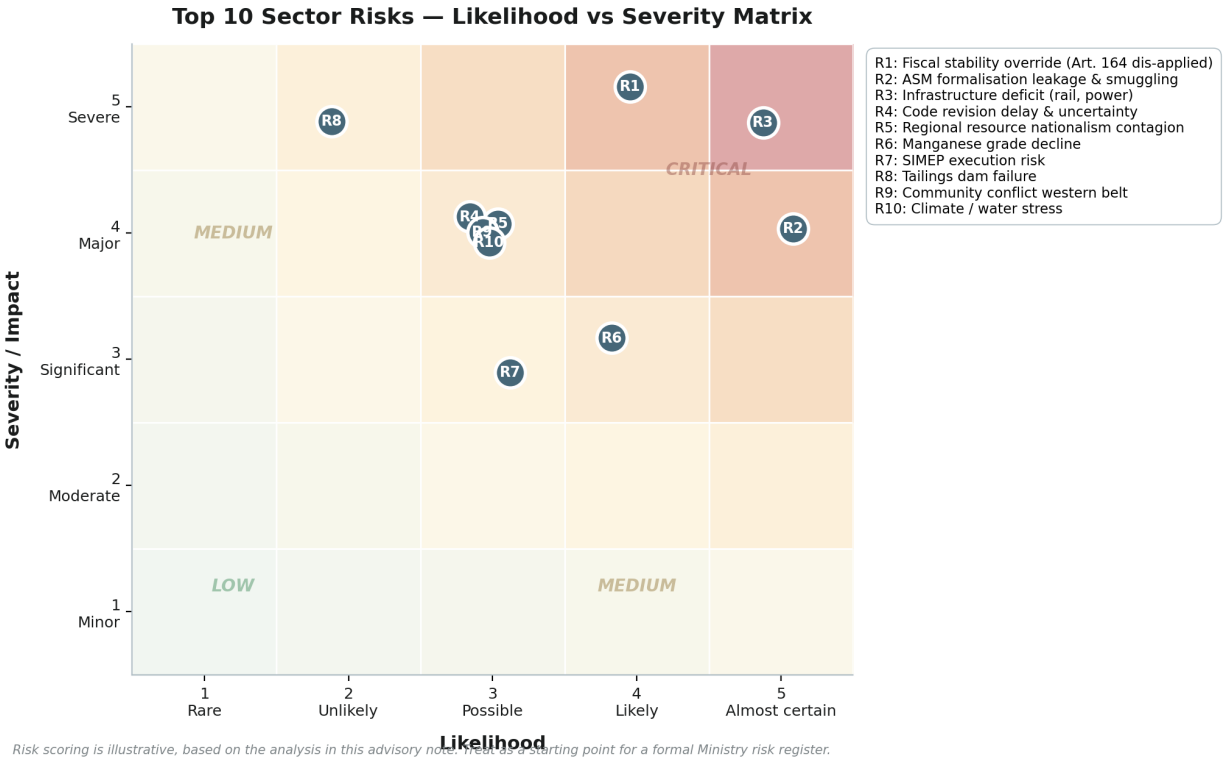


Figure 4.1: Top 10 sector risks plotted on a likelihood-severity matrix. Scoring is illustrative and should be validated by the Ministry's risk register.

4.2 Environmental Risks

The environmental risks centre on five issues. Tailings management is the highest-severity low-likelihood risk: a tailings-dam failure would cause loss of life, long-term environmental damage and reputational harm that would set the sector back by a decade. The GISTM voluntary regime (see §2.5) means that not all operators meet the same standard. Water management is a persistent concern, particularly the cross-border Bia/Tanoé river pollution from Ghanaian galamsey operations that has affected Ivorian communities. Mercury use in artisanal gold processing remains widespread and is the single largest source of mercury contamination in the sector. Deforestation from ASM activity is accelerating in the western and south-eastern gold belts. Climate and water stress is a medium-term risk: changing rainfall patterns could constrain water availability for processing at the northern mines (Koné, Séguéla) by the late 2020s.

4.3 Social Risks — Community, Land Tenure and Resettlement

Community consent and resettlement are the most visible social risks. The Rural Land Code (1998) provides a partial framework for land tenure but does not fully cover customary land rights, which are the dominant form of landholding in rural Côte d'Ivoire. Women's land rights are particularly weak under customary systems, creating gender-differentiated impacts of resettlement that operators and the State frequently fail to address. Free, Prior and Informed Consent (FPIC) is not codified in Ivorian law, although it is referenced in the 2018 Investment Code and is increasingly expected by lenders and ESG-focused investors. Local employment targets have been broadly met (94% Ivorian by 2019 per the GIZ assessment) but local procurement and subcontracting remain weak, with the supply-chain value largely captured by foreign-owned service providers. The combination of weak land tenure, insufficient FPIC implementation and shallow local procurement creates a persistent community-conflict risk that has materialised at several sites in the western belt.

4.4 Governance Risks

The governance risks are anchored in the EITI candidate-status position (see §2.7), the absence of beneficial ownership disclosure, and the convention-level (rather than codified) treatment of local content. The cumulative effect is that the State has limited visibility into the ultimate ownership of mining permits, that fiscal flows cannot be fully reconciled against production, and that operators face unequal obligations depending on the negotiating strength of their conventions. The political risk dimension — while lower than in Sahel neighbours — includes the possibility that the code-revision process delivers a more burdensome regime than operators expect, prompting a re-pricing of Ivorian mining assets and a slowdown in new investment.

4.5 Cross-Border Environmental Externalities

Côte d'Ivoire and Ghana run a joint monitoring arrangement on pollution of the Bia and Tanoé rivers from illegal mining — the rivers cross the border and the pollution from Ghanaian galamsey operations has affected Ivorian communities. The arrangement is one of the few formal bilateral mechanisms on ASM in the region, but it lacks a real-time monitoring backbone and is largely reactive to incidents reported by communities. The implication is that environmental damage is detected late, attribution is contested, and remediation is slow. A shared monitoring platform — proposed in Chapter 9 — would transform the arrangement from reactive to preventive.

Table 4.1 — Stakeholder Impact Summary by Risk Category

Risk Category	Operators	Communities	State / Government	Donors / DFIs
Fiscal-regulatory	Higher discount rate; potential investment slowdown	Indirect (lower public revenue)	Higher short-term revenue; risk of long-term investment loss	Concern over stability; possible programmatic response
Operational (tailings, water)	Compliance cost; liability exposure	Health, livelihoods, displacement	Contingent liability; reputational risk	Possible suspension of financing
ASM formalisation	ASM competition for surface rights	ASM livelihoods at risk	Revenue leakage; environmental damage	ASM formalisation programmes
Community & land tenure	Operational delays; conflict	Displacement; loss of livelihoods	Political risk; conflict mediation	FPIC and resettlement standards
Cross-border pollution	Reputational spillover	Health; loss of fisheries/agriculture	Bilateral friction; remediation cost	Regional environmental programme

Table 4.1: Stakeholder impact mapping across five risk categories.

5. Technology & AI Framework for the Government

The technology recommendations in this advisory note are organised within a single architecture: a three-layer governance stack that can be built incrementally and that supports all of the specific use cases in Chapters 6 through 9. The architecture is vendor-neutral and based on open standards, ensuring that the Government is not locked into any single supplier and that data sovereignty is preserved. This chapter describes the framework; subsequent chapters describe the two best-fit options for each priority area.

5.1 The Three-Layer Governance Stack

The recommended architecture has three layers. The **data layer** is the foundation — a National Mining Data Platform that consolidates cadastre, production, fiscal, environmental and satellite data into a single source of truth. Without this layer, every subsequent application must reconcile data silos, which is the principal cause of failed government technology projects in the mining sector globally. The **analytics layer** sits on top — machine-learning and statistical models that detect anomalies, predict outcomes and generate insights. The **decision layer** is the user-facing layer — dashboards, alerts, audit trails and reporting — that converts analytical output into decisions by the Ministry, DGMG, SODEMI and the Council of Ministers.

LAYER 1 — DATA FOUNDATION

National Mining Data Platform

Single source of truth for cadastre, production, fiscal, environmental and satellite data. Open standards (OGC, ISO 19115), sovereign data residency, modular procurement. Without this layer, all upper-layer applications must reconcile silos — the principal cause of failed government mining-tech projects globally.

5.2 Vendor-Neutral Architecture Principles

Five architecture principles guide all technology decisions. **Open standards** — Open Geospatial Consortium (OGC) for spatial data, ISO 19115 for metadata, OpenAPI for service interfaces — ensure interoperability and prevent lock-in. **Modular procurement** means each component can be replaced independently, allowing the Government to upgrade individual capabilities without rebuilding the whole stack. **Sovereign data residency** means data is stored on national infrastructure (or a regional West African cloud) and is subject to Ivorian law, with cross-border transfer governed by explicit data-sharing agreements. **Security by design** includes encryption at rest and in transit, role-based access control, and immutable audit trails for all sensitive operations. **Capacity building** is built into every contract — the supplier must train Ivorian operators and transfer knowledge, with explicit handover milestones.

5.3 Two Cross-Cutting Technologies

Two technologies are cross-cutting enablers that should be procured first and that support all subsequent use cases. The first is the National Mining Data Platform itself (Layer 1) — a single source of truth for the entire sector. The second is an AI/ML analytics layer that sits on top of the data platform and provides reusable services — anomaly detection, predictive modelling, natural-language processing — that each specific use case (Chapters 6–9) can call upon. These two technologies are pre-requisites for the phased roadmap in Chapter 11.

6. Regulatory Compliance Technology Options

This chapter addresses the regulatory gaps identified in Chapter 2 with two best-fit technology options for each priority concern. The options are vendor-neutral: each is described by capability, implementation approach and expected benefit. Procurement would be via an RFP process benchmarked against comparable deployments in Ghana, Mongolia, Peru and South Africa.

6.1 Closure Bonding & Fiscal Assurance

The gap: no codified closure-bonding methodology (§2.2). Two options address this. **Option A — Digital closure-bond calculator with bank guarantees.** A rules-based calculator that estimates the closure liability for each mine based on reserves, throughput, site footprint and rehabilitation plan, then issues a bond requirement that the operator must post as a bank guarantee held in escrow by the Treasury. Implementation: 6–9 months; capex USD 0.4–0.8 million; opex USD 80k–150k/year. Benefits: codifies the methodology, removes negotiation discretion, eliminates State contingent liability. **Option B — Independent escrow & surety-bond market platform with risk-based pricing.** A platform that allows operators to post closure bonds via independent surety providers (insurance companies) at risk-based prices, with the State holding a senior claim on the escrow account. Implementation: 12–18 months; capex USD 0.8–1.5 million; opex USD 150k–300k/year. Benefits: lower cost of capital for operators (insurance pricing vs. bank guarantee pricing), risk-based pricing that rewards good operators, deeper domestic insurance market.

6.2 Beneficial Ownership Disclosure

The gap: no UBO disclosure regime (§2.7). Two options. **Option A — UBO register integrated with EITI reporting.** A central register of ultimate beneficial owners of all mining permit holders, integrated with EITI reporting to meet the 2023 EITI Standard. Operators self-declare; the Ministry verifies against corporate registries. Implementation: 12–15 months; capex USD 0.5–1.0 million; opex USD 100k–200k/year. Benefits: EITI compliant-status achievement, reduced corruption risk, investor confidence. **Option B — Graph AI for ownership-network analysis and Politically Exposed Persons (PEP) detection.** A graph database that maps the full ownership network (parent companies, subsidiaries, directors, beneficial owners) and applies AI to flag PEPs, sanctioned entities and suspicious ownership patterns. Implementation: 18–24 months (including Option A as foundation); capex USD 1.2–2.0 million; opex USD 250k–400k/year. Benefits: proactive detection of illicit financial flows, ability to meet FATF Recommendation 24, regional leadership on beneficial-ownership transparency.

6.3 Permit & e-Cadastre Upgrade

The gap: e-cadastre exists since September 2025 but is a thin portal without SLA tracking, inter-agency integration or public transparency. Two options. **Option A — One-stop digital permitting portal with SLA tracking.** Extends the current e-cadastre into a one-stop portal that integrates with the Environment Ministry (ESIA), Water Directorate, Forestry, Customs, Tax and SODEMI, with statutory SLA tracking and automatic escalation when deadlines are missed. Implementation: 12 months; capex USD 0.8–1.5 million; opex USD 150k–250k/year. Benefits: faster permit decisions, reduced corruption risk (transparent SLAs), better investor experience. **Option B — Blockchain-based mining-title registry for immutable title history.** A permissioned blockchain (Hyperledger Fabric or equivalent) that records every permit grant, transfer, encumbrance and revocation as an immutable transaction, with public transparency for non-confidential fields. Implementation: 18–24 months; capex USD 1.5–2.5 million; opex USD 300k–500k/year. Benefits: complete title-history audit trail, tamper-evident record, public confidence in title integrity, alignment with the African Union's Africa Mining Vision transparency objectives.

Table 6.1 — Comparison of Regulatory Compliance Technology Options

Use Case	Option	Indicative Capex (USD m)	Indicative Opex (USD m/yr)	Timeline	Primary Benefit
Closure bonding	A: Calculator + bank guarantees	0.4–0.8	0.08–0.15	6–9 months	Codifies methodology; eliminates State liability
Closure bonding	B: Surety-bond market platform	0.8–1.5	0.15–0.30	12–18 months	Risk-based pricing; lower operator cost of capital
Beneficial ownership	A: UBO register + EITI	0.5–1.0	0.10–0.20	12–15 months	EITI compliance; reduced corruption risk
Beneficial ownership	B: Graph AI + PEP detection	1.2–2.0	0.25–0.40	18–24 months	Proactive illicit-flow detection; FATF R.24
Permit / e-cadastre	A: One-stop portal + SLAs	0.8–1.5	0.15–0.25	12 months	Faster decisions; transparent SLAs
Permit / e-cadastre	B: Blockchain title registry	1.5–2.5	0.30–0.50	18–24 months	Immutable title history; tamper-evident

Table 6.1: Side-by-side comparison of the two regulatory compliance technology options per use case. Indicative figures based on comparable deployments.

7. Infrastructure Monitoring Technology Options

The infrastructure gap (Chapter 3) cannot be closed by technology alone — rail and port investment require capital expenditure in the billions of dollars. But technology can address two related problems: revenue protection (ensuring that the State captures the full royalty and tax due from bulk-commodity exports that move through existing infrastructure) and operational optimisation (ensuring that existing infrastructure is used efficiently). This chapter addresses two priority use cases: bulk-commodity logistics monitoring, and power and water monitoring for mines.

7.1 Bulk-Commodity Logistics Monitoring

Option A — IoT weigh-in-motion sensors at mine gates + rail/port integration. Weigh-in-motion (WIM) sensors installed at mine gates and at port entry points capture the weight of every truck leaving a mine and every truck entering a port, automatically reconciling against operator-declared production and royalty declarations. Discrepancies trigger automatic alerts to DGMG. Implementation: 9–12 months; capex USD 1.0–2.0 million (covering 12–20 sites); opex USD 200k–400k/year. Benefits: estimated 5–10% revenue uplift on manganese royalties through under-declaration detection, real-time production visibility, anti-smuggling deterrent effect. **Option B — Satellite-based stockpile and shipment monitoring (optical + SAR).** Satellite imagery (Sentinel-2 optical, Sentinel-1 SAR, plus commercial high-resolution where needed) is used to estimate stockpile volumes at mines and ports, and to track vessel loading at port. AI models compare satellite-estimated volumes against operator-declared volumes. Implementation: 6–9 months for the analytics platform; capex USD 1.2–2.5 million; opex USD 250k–500k/year (including imagery licensing). Benefits: independent verification of operator declarations, coverage of remote sites without ground sensors, ability to detect undeclared production.

7.2 Power and Water Monitoring for Mines

Option A — Smart-grid monitoring with mine-level meters. Smart meters installed at mine substations provide real-time power-consumption data, allowing the grid operator and the Ministry to verify mine production against power consumption (a standard indicator of throughput). Implementation: 9–12 months; capex USD 0.6–1.2 million; opex USD 100k–200k/year. Benefits: production verification, grid management improvement, ability to detect undeclared processing. **Option B — Hydrological sensors and water-stress early-warning system for watersheds.** A network of hydrological sensors in the Bandama, Bia and Tanoé watersheds, with AI-based modelling of water-stress scenarios under different climate and demand projections. Implementation: 12–18 months; capex USD 0.7–1.5 million; opex USD 150k–300k/year. Benefits: early warning of water-stress constraints on mine operations, cross-border pollution detection (Bia/Tanoé), environmental-flow protection.

Table 7.1 — Comparison of Infrastructure Monitoring Technology Options

Use Case	Option	Indicative Capex (USD m)	Indicative Opex (USD m/yr)	Timeline	Primary Benefit
Bulk logistics	A: IoT WIM + rail/port integration	1.0–2.0	0.20–0.40	9–12 months	5–10% royalty uplift; anti-smuggling
Bulk logistics	B: Satellite stockpile monitoring	1.2–2.5	0.25–0.50	6–9 months	Independent verification; remote-site coverage
Power	A: Smart-grid + mine-level meters	0.6–1.2	0.10–0.20	9–12 months	Production verification; grid management
Water	B: Hydrological sensors + early warning	0.7–1.5	0.15–0.30	12–18 months	Water-stress early warning; cross-border pollution

Table 7.1: Side-by-side comparison of infrastructure monitoring technology options.

8. ASM Detection & Formalisation Technology

The ASM gap (§2.3) is the highest-priority technology use case for the Government. The estimated 5–15 tonnes per year of informal gold flow represents both a revenue loss (royalty and tax not captured) and an ESG externality (mercury use, deforestation, river pollution). Two technology options address the gap from complementary angles: detection (identifying informal ASM activity) and formalisation (bringing ASM operators into the licensed, traceable system).

8.1 Option A — Satellite + AI Illegal-Mining Detection Platform

A platform that combines multi-spectral satellite imagery (Sentinel-2 for routine coverage, commercial high-resolution for targeted inspection) with machine-learning models that detect ASM activity through three signals: deforestation patterns (vegetation loss in forested areas), water turbidity changes (sediment plumes in rivers downstream of ASM activity), and spectral signatures of excavation. Alerts are issued to DGMG field teams for ground verification. Implementation: 6–9 months for the analytics platform, 12 months for full national coverage; capex USD 1.2–2.5 million; opex USD 250k–500k/year (including imagery licensing). Benefits: detection of previously unknown ASM sites, ability to monitor remote areas without ground patrols, evidence base for enforcement actions, environmental remediation prioritisation. The platform can also support the joint Bia/Tanoé monitoring arrangement with Ghana (§4.5) by extending the analytical coverage across the border.

8.2 Option B — Mobile-Based ASM Registration and Gold Traceability

A mobile application that allows ASM operators to register for an artisanal mining licence, declare production, and sell gold through formal channels — including the SIMEP refinery (post-H1 2027). The app uses a digital ID for each ASM operator and creates a chain-of-custody record for gold from pit to refinery, with QR-code verification at each transfer. Implementation: 12–18 months; capex USD 1.0–2.0 million; opex USD 200k–400k/year. Benefits: formalisation rate increase (target 30–50% of ASM production within 3 years), revenue capture (royalty and tax on previously informal gold), environmental improvement (formal operators are subject to mercury-use restrictions and rehabilitation requirements), social benefit (formal ASM operators access fair pricing through SIMEP). The traceability chain also supports OECD Due Diligence Guidance for responsible mineral supply chains.

Table 8.1 — Comparison of ASM Detection and Formalisation Technology Options

Option	Approach	Indicative Capex (USD m)	Indicative Opex (USD m/yr)	Timeline	Primary Benefits
A: Satellite + AI detection	Multi-spectral satellite + ML detection of deforestation, water turbidity, excavation	1.2–2.5	0.25–0.50	6–12 months	Remote detection; enforcement evidence base; environmental remediation prioritisation
B: Mobile ASM registration + traceability	Mobile app with digital ID, production declarations, SIMEP chain-of-custody	1.0–2.0	0.20–0.40	12–18 months	30–50% formalisation in 3 years; revenue capture; OECD Due Diligence compliance

Table 8.1: Two complementary ASM technology options — Option A detects informal activity, Option B formalises the operators.

9. ESG & Environmental Monitoring Technology

ESG monitoring technology addresses the environmental risks identified in §4.2 — tailings dam safety, water quality, and land-use change. The two priority use cases are tailings dam safety monitoring (the highest-severity risk in the matrix) and environmental compliance monitoring across watersheds and mining regions.

9.1 Tailings Dam Safety Monitoring

Option A — IoT geotechnical sensors + AI anomaly detection. A network of piezometers (pore-water pressure), inclinometers (slope movement), and settlement sensors installed on priority tailings dams, with data telemetered to a real-time dashboard and AI models that flag anomalies against baseline behaviour. Alerts are sent to the operator, DGMG and the Ministry. Implementation: 12–15 months for the first 5 priority dams; capex USD 1.5–3.0 million; opex USD 300k–600k/year. Benefits: early warning of structural failure, regulatory compliance with GISTM (§2.5), ESG reporting improvement, reduced insurance premiums for operators. **Option B — Drone + satellite InSAR deformation monitoring.** Satellite SAR (Sentinel-1, plus commercial X-band for higher resolution) processed via Interferometric Synthetic Aperture Radar (InSAR) to detect millimetre-level surface deformation on and around tailings dams, complemented by quarterly drone surveys for high-resolution visual inspection. Implementation: 6–9 months for the analytics platform; capex USD 1.0–2.0 million; opex USD 200k–400k/year. Benefits: independent verification of operator-reported monitoring data, coverage of all dams (not just instrumented ones), early detection of deformation patterns that precede failure.

9.2 Environmental Compliance Monitoring

Option A — Multi-sensor water quality network (Bia/Tanoé cross-border with Ghana). A network of water-quality sensors (pH, turbidity, conductivity, dissolved oxygen, heavy-metals) at strategic points on the Bia and Tanoé rivers, with data shared in real time between the Ivorian and Ghanaian authorities under the joint monitoring arrangement (§4.5). Implementation: 9–12 months; capex USD 0.5–1.2 million; opex USD 100k–250k/year. Benefits: real-time cross-border pollution detection, evidence base for remediation claims, transformation of the bilateral arrangement from reactive to preventive. **Option B — AI-powered EIA monitoring with satellite land-use change verification.** AI models that compare satellite-observed land-use change against the environmental impact assessment (EIA) commitments made by each operator, flagging discrepancies for DGMG inspection. Implementation: 12–15 months; capex USD 0.8–1.5 million; opex USD 150k–300k/year. Benefits: independent verification of EIA compliance, ability to prioritise inspections, support for EITI environmental disclosure requirements.

Table 9.1 — Comparison of ESG and Environmental Monitoring Technology Options

Use Case	Option	Indicative Capex (USD m)	Indicative Opex (USD m/yr)	Timeline	Primary Benefit
Tailings safety	A: IoT geotechnical sensors + AI	1.5–3.0	0.30–0.60	12–15 months	Early warning; GISTM compliance
Tailings safety	B: Drone + satellite InSAR	1.0–2.0	0.20–0.40	6–9 months	Independent verification; full dam coverage
Water quality	A: Multi-sensor Bia/Tanoé network	0.5–1.2	0.10–0.25	9–12 months	Cross-border pollution detection
EIA compliance	B: AI land-use verification	0.8–1.5	0.15–0.30	12–15 months	Independent EIA compliance verification

Table 9.1: ESG and environmental monitoring technology options.

10. Process Improvement Technologies

Beyond the use-case-specific technologies in Chapters 6–9, five cross-cutting technologies can improve Ministry processes across the board. These technologies typically deliver cost savings and efficiency gains rather than direct revenue uplift, but their cumulative effect on Ministry capacity is material — particularly at a time when permit-grant volume is rising (151 of 189 in 2023; 171 of 225 in 2025) and the code-revision process will add new regulatory obligations.

10.1 Robotic Process Automation (RPA)

RPA automates repetitive, rules-based tasks: permit-application validation, royalty-declaration reconciliation, EITI report compilation, inter-agency data exchange. Implementation: 6–9 months for the first three use cases; capex USD 0.3–0.6 million; opex USD 80k–150k/year. Benefits: 40–60% reduction in processing time for routine tasks, freeing Ministry staff for higher-value activities (inspection, audit, policy analysis). RPA is the lowest-hanging fruit and should be procured early to demonstrate quick wins and build momentum for the broader technology programme.

10.2 Predictive Analytics for Production Forecasting and Revenue Modelling

Machine-learning models that forecast national production by commodity (gold, manganese, nickel) based on operator guidance, satellite-observed activity, and macroeconomic scenarios, and that model the fiscal implications under different royalty and tax scenarios. Implementation: 9–12 months; capex USD 0.5–1.0 million; opex USD 100k–200k/year. Benefits: improved budget planning, ability to model the fiscal impact of code-revision options, support for the SIMEP refinery throughput planning.

10.3 Digital Twin for Mine-Site Planning and Infrastructure Corridor Design

A digital twin — a 3D + temporal model of mine sites, infrastructure corridors (rail, road, power, port) and watersheds — that allows the Ministry and operators to model the impact of infrastructure investment decisions before committing capital. The twin is particularly relevant for evaluating the proposed rail extensions to the western iron-ore belt (§3.2). Implementation: 18–24 months for the first version; capex USD 2.5–5.0 million; opex USD 500k–1 million/year. Benefits: de-risked infrastructure investment, ability to attract blended finance (public-private partnership) by quantifying returns, support for PIRME 2026-2040 planning.

10.4 Generative AI for Ministerial Briefings, Contract Review and Citizen Queries

A retrieval-augmented generation (RAG) system that grounds a large-language model on the Mining Code, implementing decrees, mining conventions (non-confidential clauses), EITI reports, and prior Ministry decisions. The system drafts ministerial briefings, reviews operator-submitted documents for compliance, and answers citizen queries via a public portal. Implementation: 6–9 months for an internal-only pilot; 12–15 months for public-facing deployment; capex USD 0.5–1.2 million; opex USD 150k–300k/year. Benefits: 30–50% reduction in staff time on routine briefings and reviews, consistent application of regulatory interpretation, improved citizen engagement.

10.5 Open Data Portal for Transparency

A public open data portal publishing non-confidential mining data: permit boundaries, production statistics (aggregated to operator level), royalty and tax payments, EITI reports, environmental monitoring summaries. Implementation: 6–9 months; capex USD 0.4–0.8 million; opex USD 80k–150k/year. Benefits: EITI compliant-status achievement, reduced information asymmetry for investors and communities, support for academic and civil-society analysis that strengthens the policy ecosystem.

11. Implementation Roadmap & ROI

This chapter consolidates the technology recommendations into a phased 2026–2030 roadmap, with indicative capex and opex bands and an illustrative ROI model. The roadmap is structured in three phases: foundations (2026), scaling (2027–2028), and advanced analytics (2029–2030). The phasing reflects dependencies (the National Mining Data Platform must exist before upper-layer applications can be built) and capacity constraints (the Ministry and DGMG cannot absorb the full technology stack simultaneously).

11.1 Phased Roadmap 2026–2030

Phase 1 (2026) builds the foundations: the National Mining Data Platform, an upgraded e-cadastre portal, an ASM satellite-detection pilot in two regions, and the design of a UBO register. The phase also includes the publication of the gap analysis (this advisory note) and code-revision principles, ensuring that the technology programme and the regulatory reform process are sequenced together. Phase 2 (2027–2028) scales the foundations to national coverage: full ASM detection, UBO register live, tailings IoT sensors on priority dams, SIMEP gold traceability integration, hydrological sensors on the Bia/Tanoé, satellite stockpile monitoring, and the target enactment of the revised Mining Code. Phase 3 (2029–2030) layers advanced analytics: AI fiscal anomaly detection, generative AI for policy briefings, a digital twin for mine-site and infrastructure corridor planning, an open data portal, and an independent evaluation of the programme to capture lessons learned.

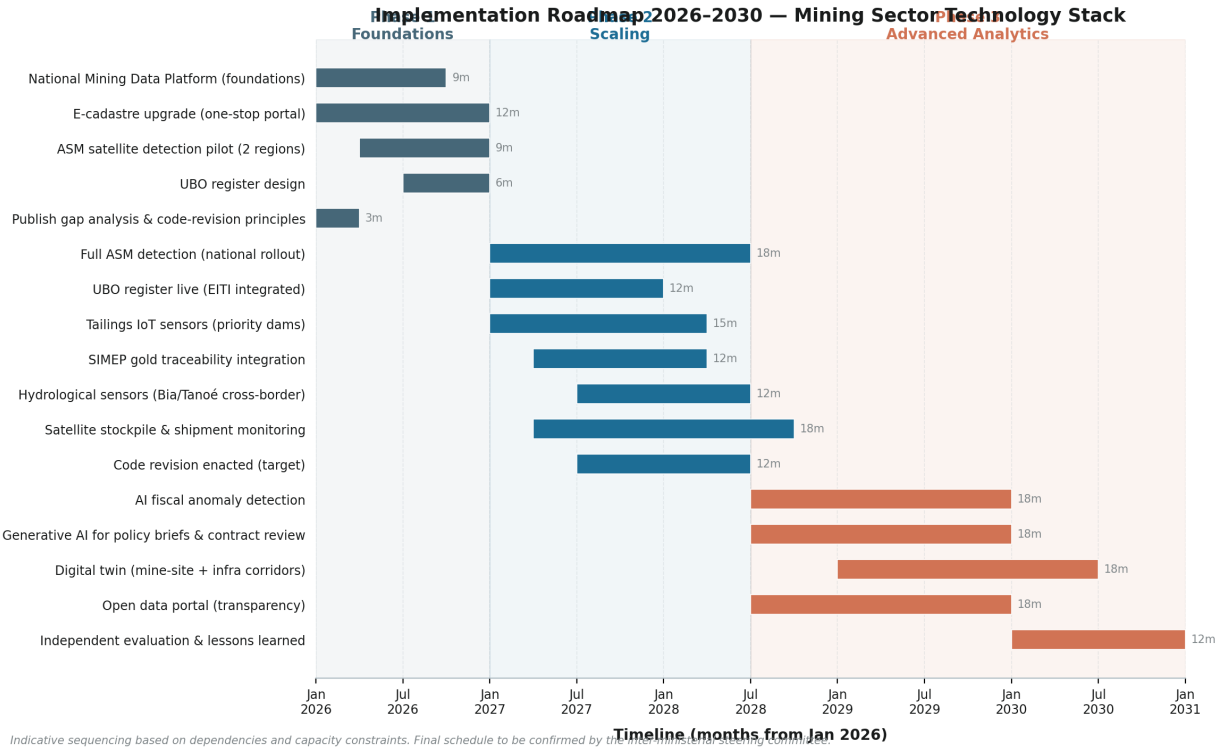


Figure 11.1: Implementation roadmap 2026–2030 — phased technology deployment across three horizons.

11.2 Investment Bands and Indicative Capex

The chart below shows indicative capex (one-off) and annual benefit (recurring) for each technology in the stack. Capex bands are based on comparable deployments in Ghana, Mongolia, Peru and South Africa, adjusted for Ivorian cost factors. Annual benefit combines revenue uplift (royalty and tax capture), cost savings (Ministry operating efficiency), and avoided losses (environmental remediation, contingent liability). Payback periods are annotated on the chart: green indicates sub-one-year payback, amber indicates one-to-three years, red indicates over three years. The technologies with the shortest payback (e-cadastre upgrade, ASM satellite detection, AI fiscal anomaly detection, satellite stockpile monitoring) should be prioritised within Phase 1 and Phase 2.

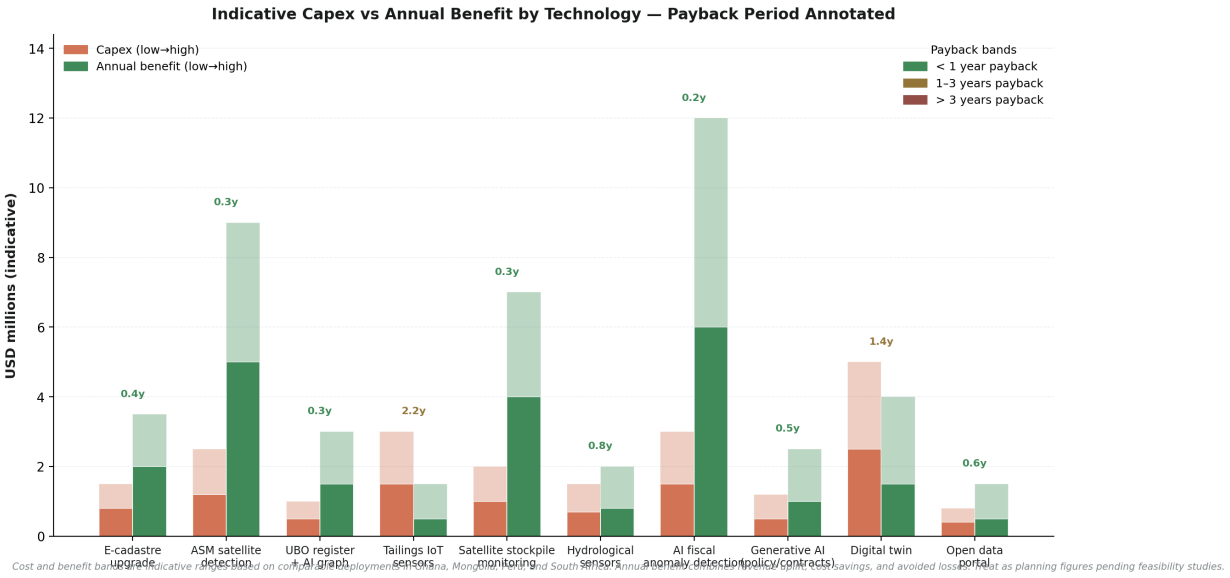


Figure 11.2: Indicative capex vs annual benefit by technology, with payback period annotations. Treat as planning figures pending feasibility studies.

11.3 ROI Model — Aggregate View

The aggregate capex across the full technology stack is in the USD 10–25 million range over the 2026–2030 horizon, with recurring opex of USD 2–4 million per year by 2030. The aggregate annual benefit (combining revenue uplift, cost savings and avoided losses) is in the USD 22–45 million per year range by 2030, suggesting a programme-level payback period of 12–24 months once the stack is fully deployed. The largest benefit contributors are: ASM satellite detection (revenue uplift on informal gold), AI fiscal anomaly detection (royalty and tax recovery), satellite stockpile monitoring (revenue protection on bulk commodities), and the e-cadastre upgrade (efficiency and corruption-risk reduction).

Table 11.1 — Aggregate Investment and Benefit Summary by Phase

Phase	Period	Cumulative Capex (USD m)	Annual Opex by Phase End (USD m)	Annual Benefit by Phase End (USD m)	Indicative Payback
Phase 1 — Foundations	2026	3.0–5.5	0.6–1.0	5.0–9.0	8–12 months
Phase 2 — Scaling	2027–2028	8.0–16.0	1.6–2.8	16.0–28.0	12–18 months
Phase 3 — Advanced	2029–2030	10.0–25.0	2.4–4.5	22.0–45.0	12–24 months

Table 11.1: Aggregate investment and benefit summary by phase. Figures are indicative ranges.

11.4 Governance and Capacity Building

The technology programme requires governance and capacity-building investments that are easy to underestimate. An inter-ministerial steering committee (Mines, Environment, Finance, Digital Economy, SODEMI) should oversee the programme, with a dedicated programme office within the Ministry of Mines. Vendor management should follow a modular-procurement model: no single supplier delivers more than two components, and each contract includes knowledge-transfer milestones with explicit Ivorian-staff succession plans. Sovereign data residency is non-negotiable: all data must be stored on national or regional (West African) infrastructure, with cross-border transfer governed by explicit data-sharing agreements. Training is the most under-budgeted line item globally — at least 15% of capex should be reserved for Ministry staff training and ongoing skills development.

12. Recommendations & Next Steps

This chapter consolidates the analysis into five priority recommendations, a 90-day quick-win agenda, and a 12-month action plan. The recommendations are sequenced to deliver early wins (build confidence in the programme), address the highest-priority gaps (ASM formalisation, fiscal stability, closure bonding) and lay the foundations for the longer-term technology build-out.

12.1 Top Five Recommendations

Recommendation 1 — Establish the National Mining Data Platform. The data platform is the foundation for everything else. Without it, every subsequent application must reconcile data silos, which is the principal cause of failed government mining-tech projects globally. Procurement should begin in Q1 2026, with a target go-live in Q4 2026. **Recommendation 2 — Pilot ASM satellite detection in two regions.** ASM formalisation is the highest-priority gap (§2.3, §4.2). A satellite-detection pilot in the western and south-eastern belts, combined with the mobile-based ASM registration and traceability app, can demonstrate measurable revenue capture and environmental improvement within 12 months. **Recommendation 3 — Publish a UBO roadmap and begin register design.** The EITI compliant-status achievement requires beneficial-ownership disclosure. A public roadmap published in Q1 2026, followed by register design in Q2–Q4 2026 and go-live in Q1–Q2 2027, would position Côte d'Ivoire as a regional leader. **Recommendation 4 — Codify closure bonding in the revised Mining Code.** The closure-bonding gap (§2.2) exposes the State to contingent liability that grows as the first-wave mines approach end-of-life. The revised code should mandate risk-based closure bonds calculated by independent third parties and held in escrow. **Recommendation 5 — Establish the inter-ministerial steering committee.** The technology programme requires cross-ministerial coordination that no single ministry can deliver alone. A steering committee chaired by the Ministry of Mines, with Environment, Finance, Digital Economy and SODEMI as members, should be established by Q1 2026.

12.2 90-Day Quick Wins (Q1 2026)

Three actions can be delivered within 90 days and would build confidence in the broader programme. First, publish the gap analysis and code-revision principles — this advisory note can serve as the basis, with Ministry adaptation. Second, launch the ASM satellite-detection pilot in two regions (western and south-eastern gold belts) — a 6-month pilot would deliver measurable results by end-Q2 2026. Third, publish the UBO roadmap with explicit milestones — this signals intent to investors, donors and EITI without committing to specific technology procurement.

12.3 12-Month Action Plan (2026)

By end-2026, the Ministry should have: (i) the National Mining Data Platform in pilot operation; (ii) the e-cadastre one-stop portal in design, with procurement launched; (iii) the ASM satellite-detection pilot completed and scaled to national procurement decision; (iv) the UBO register in design, with go-live targeted for Q1 2027; (v) the inter-ministerial steering committee operational with a published programme charter; (vi) the revised Mining Code principles published, with stakeholder consultation underway; and (vii) the first RPA use cases deployed (permit-application validation, EITI report compilation).

12.4 Long-Term Institutional Reforms (2027–2030)

Beyond the immediate technology programme, three institutional reforms would consolidate the gains. First, establish a Mining Sector Digital Authority within or adjacent to SODEMI, with a mandate to operate the technology stack on a sustainable funding model (a small percentage of royalty revenue). Second, develop a sovereign data centre for the mining sector, ensuring that all data is stored on national infrastructure with appropriate security and disaster-recovery provisions. Third, deepen regional cooperation with Ghana on ASM monitoring — the joint Bia/Tanoé arrangement is a foundation that can be extended to a regional ASM monitoring platform and eventually to a regional gold-traceability system aligned with the OECD Due Diligence Guidance. These reforms position Côte d'Ivoire as a regional leader on mining governance and create the institutional infrastructure that will outlast any individual technology deployment.

12.5 Cross-Reference to the Prior Sector Note

This advisory note supplements the *Côte d'Ivoire Mining & Minerals Sector Note* (October 2026), which provides the sector baseline — output trajectory, deposits, operators, project pipeline, regulatory framework, regional context. The prior note's Section 7 (Regulatory Overview) provides the underlying legal detail for Chapter 2 of this note; the prior note's Section 4 (Infrastructure & Logistics) provides the underlying sectoral detail for Chapter 3 of this note; and the prior note's Section 8 (Key Takeaways, Risks & Outlook) provides the sectoral risk identification that Chapter 4 of this note operationalises into a risk matrix. The two notes are designed to be read together.

Appendix — Sources, Caveats & Cross-Reference

A.1 Methodology

This advisory note was compiled from the prior sector note (*Côte d'Ivoire Mining & Minerals Sector Note*, October 2026) supplemented by public sources including the USGS National Minerals Information Center, the World Bank's Africa Mining Vision documentation, the EITI Standard 2023 and country reports, UNECA ASM profiles, OECD Due Diligence Guidance, the GISTM standard, and comparable African technology deployments in Ghana, South Africa, Mongolia and Peru. The vendor-neutral stance means that no specific product or supplier is recommended; all cost bands are indicative ranges based on comparable deployments and require feasibility confirmation before procurement.

A.2 Caveats

Three caveats apply. First, cost and benefit bands are indicative ranges based on comparable deployments; actual costs will depend on the scope of each procurement, the supplier market at the time, and the integration complexity with existing Ministry systems. Second, the risk matrix scoring (Chapter 4) is illustrative and should be validated by the Ministry's own risk register, ideally through a structured expert-elicitation process. Third, the technology options presented in Chapters 6–9 are the two best-fit options per use case identified by the analyst; alternative options exist and the procurement process should include an open RFP that allows suppliers to propose alternative approaches.

A.3 Standards Referenced

The following standards and frameworks are referenced in this advisory note: the EITI Standard 2023 (beneficial ownership disclosure, contract transparency, revenue reconciliation); the Global Industry Standard on Tailings Management (GISTM, 2020); the UEMOA Community Mining Code (Regulation 02/2023/CM/UEMOA); the OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas (3rd edition, 2016); the Initiative for Responsible Mining Assurance (IRMA) Standard; the FATF Recommendations (specifically Recommendation 24 on beneficial ownership); and the Africa Mining Vision (2009). The technology architecture recommendations in Chapter 5 align with the Open Geospatial Consortium (OGC) standards for spatial data and ISO 19115 for metadata.

A.4 Cross-Reference

This advisory note is designed to be read alongside the *Côte d'Ivoire Mining & Minerals Sector Note* (October 2026). The prior note provides the sector baseline; this note provides the governance, risk and technology overlay. Together they constitute a comprehensive view of the Ivorian mining sector as of October 2026, suitable for Ministry planning, donor engagement and investor briefing.